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Asthma: Clinical Findings and Diagnosis 2026

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November 2026

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Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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- Define and recognize the symptoms of asthma
- Understand the epidemiology of asthma and identify the different types of asthma
- Describe the relationship of IgE to asthma and distinguish the various roles that environment and genetics play in asthma
- Detect mimickers of asthma
- Recognize status asthmatics and warning signs for severe asthmatics
- Explain the clinical presentation and tests necessary to diagnose asthma and come up with a differential diagnosis
- Know the goal of asthma treatment and management

Indicates tested on the boards



BACKGROUND

"As we have become more and more dependent upon the laboratory and special investigations in making our diagnosis, we have gradually lost that faculty of clinical observation which enabled these early clinicians to make an accurate diagnosis in the more common diseases almost with a glance at the patient."

Dr. Will Mayo

History of the Diagnosis of Asthma

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Hippocrates:
Father of Medicine

Father of Medicine

460-357 BC

Recognized the spasmodic nature of asthma

Believed it to be caused by moisture, occupation and climate



Maimonides: 12th century physician

12th Century

Compiled volume on asthma: sudden bouts of dyspnea

Physician to the Sultan Saladin of Egypt

Treatment included chicken soup and sexual abstinence



Sir John Floyer:
Physician and asthmatic

Physician and patient with asthma
1698: *A Treatise of the Asthma*
Observed and commented on his symptoms:
Described influence of environment
Believed asthma due to constriction of the bronchi

History of the Diagnosis of Asthma

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- In 1907 all agreed that a neurosis is an important factor, but many believe that is also an inflammatory affection of the bronchioles that is either the primary or secondary cause of this condition
  - Clinically the disease presents in three fundamental symptoms
    1. The paroxysmal charter of the attacks
    2. The dyspnea
    3. The vasosecretory disturbance

Current Definition of Asthma

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- "Asthma is a **heterogeneous** disease, usually characterized by chronic airway inflammation. It is defined by the history of the respiratory symptoms such as wheeze, shortness of breath, chest tightness and cough that vary over time and in intensity, together with variable expiatory airflow limitation" - Global Initiative for Asthma 2017
- An obstructive lung disease that is ***reversible***

Defining Asthma

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- Symptoms that **increased** the probability for asthma diagnosis
  - More than one of the classic symptoms (wheeze, SOB, cough, chest tightness)
  - Symptoms worse and night and early morning
  - Symptoms vary over time and vary in intensity
  - Symptoms are triggered by viruses, exercise, allergens, changes in weather, laughter or irritants (smoke, strong smells, fumes)
- Symptoms that **decrease** the probability for asthma diagnosis
  - Isolated cough with no other respiratory symptoms
  - Chronic sputum production
  - SOB associated with dizziness, light-headedness or peripheral tingling
  - Chest pain
  - Exercise-induced dyspnea with noisy inspiration



Approximated Annual Effects of Asthma

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- 14.4 million school days missed from asthma
- 14.2 million work days missed
- 11.7 million asthma-related healthcare visits
- 2.1 million asthma-related emergency department visits
- 439,000 asthma-related hospitalizations
- **3,388 deaths due to asthma**

Epidemiology of Asthma

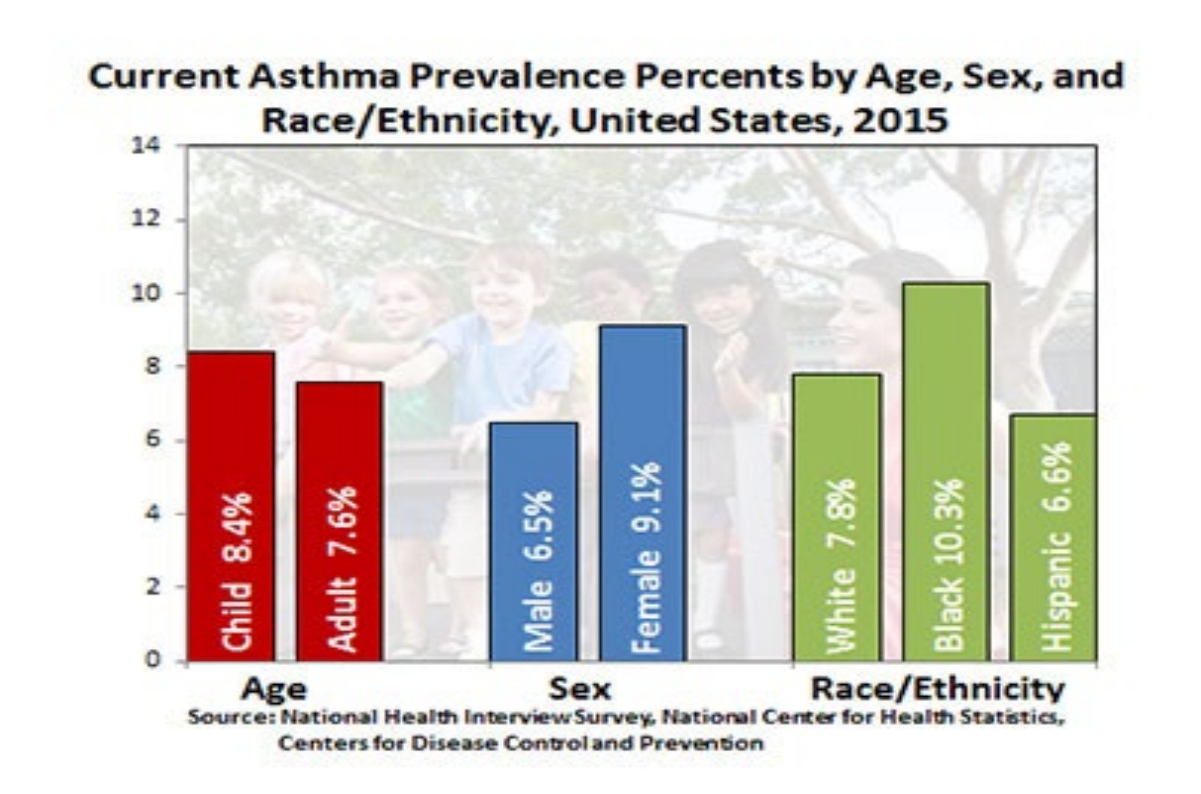
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- According to the Center for Disease Control (CDC) asthma prevalence in the U.S:
  - ↑↑ from the early 1980's to the mid 1990's
  - ↑, but more gradually than early years, from 2001- 2009
  - Stabilized from 2009 to 2012
  - ↓in 2013

Epidemiology of Asthma

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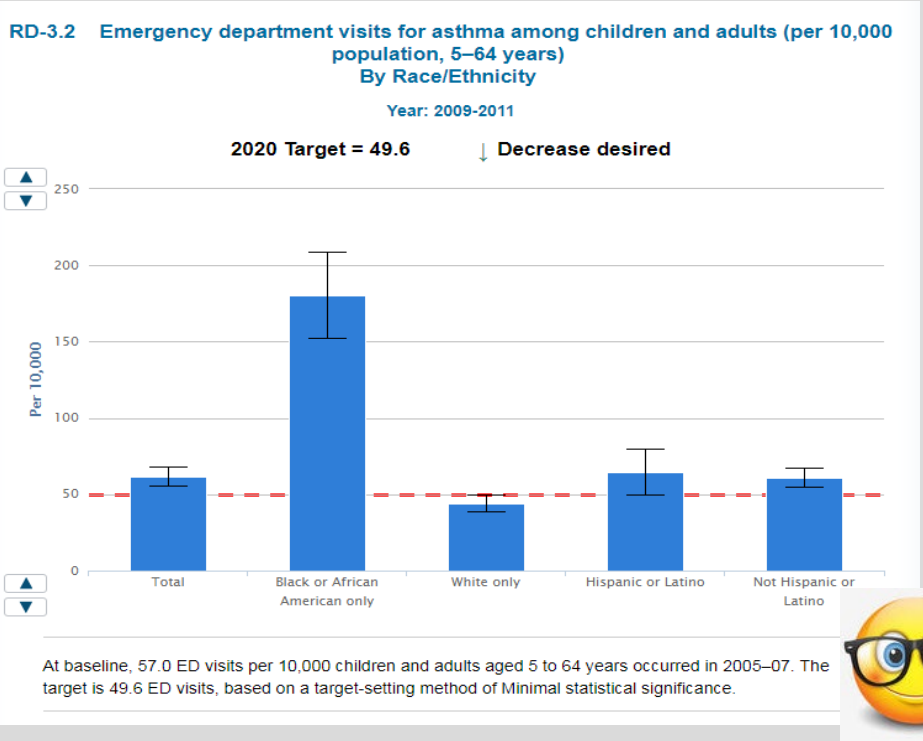
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Epidemiology of Asthma by Race/Ethnicity

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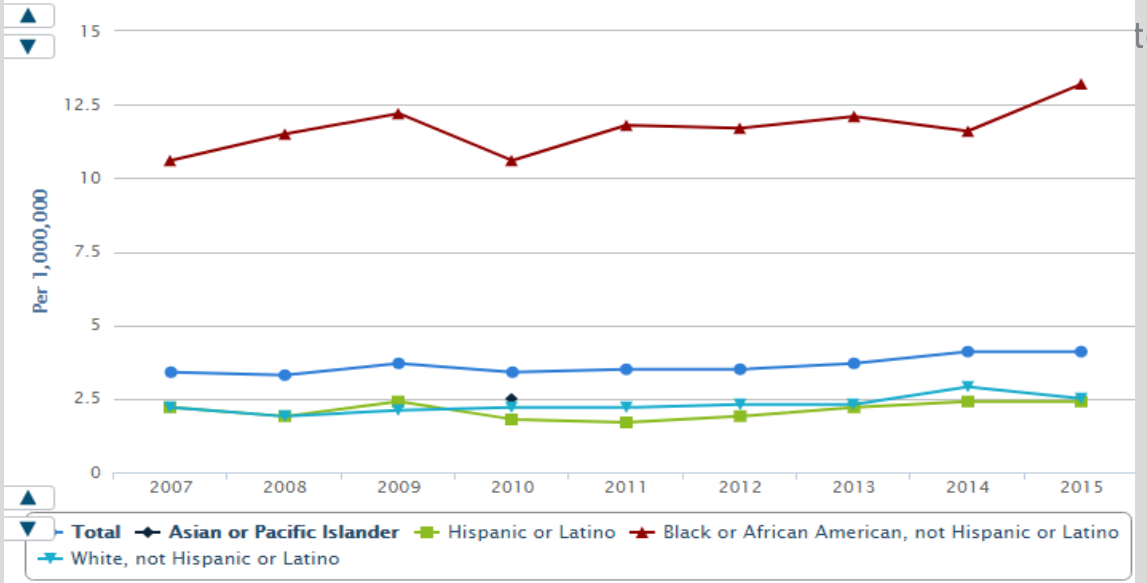
- White: 7.8%
  - Child: 7.4%
  - Adult: 7.9%
- Hispanic: 6.6%
  - Child: 8.0%
  - Adult: 5.9%
- **Black Non-Hispanic: 10.3%**
  - Child: 13.4%
  - Adult: 9.1%
- Other Non-Hispanic: 6.9%
  - Child: 8.4%
  - Adult: 6.3%





RD-1.1 Asthma deaths among children and adults (per million population, <35 years) By Race/Ethnicity

2020 Target = Not applicable ↓ **Decrease desired**



At baseline, 3.4 asthma deaths per million children and adults under age 35 years occurred in 2007. This measure is being tracked for informational purposes. If warranted, a target will be set during the decade.

Data Source: National Vital Statistics System-Mortality (NVSS-M), CDC/NCHS; Bridged-race Population Estimates, CDC/NCHS and Census

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Epidemiology of Asthma and Poverty

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- Below 100% of the poverty line: 11.1%
- 100% to less than 250% of poverty line: 8.4%
- 250% to less than 450% of poverty line: 6.3%
- 450% of the poverty line or higher: 6.9%

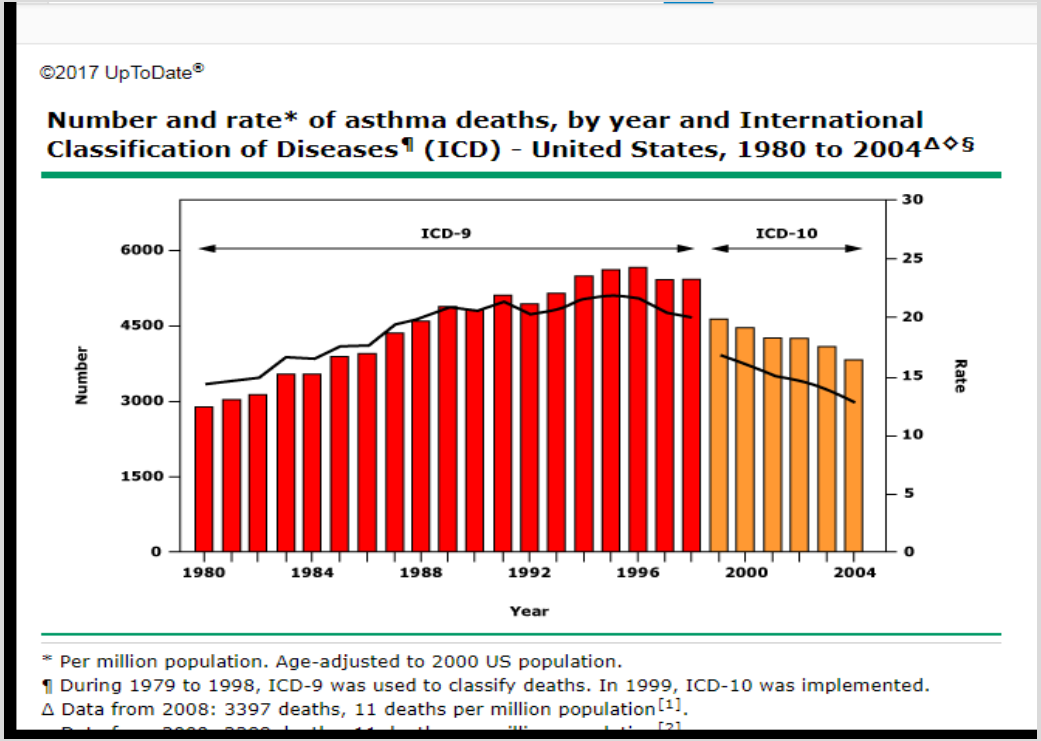
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Asthma Disparities Among US Children

- Lower-income populations, minorities and children living in inner cities experience more ED visits, hospitalizations, and death due to asthma than the general population
- The burden of asthma falls disproportionately on non-Hispanic black, American Indian/Alaskan Native and some Hispanic populations.

Epidemiology of Asthma

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ICD 9 classified asthma as intrinsic or extrinsic

ICD 10 Classifies by disease severity

CLASSIFICATION & CAUSATION

Current Classifications of Asthma

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- **Allergic (Atopic) asthma**
  - Personal and/or family history of allergic diseases
  - Atopy
    - Positive skin tests
  - Increased levels of serum IgE
  - Positive response to bronchoprovocation with specific antigen

- **Idiosyncratic asthma**
  - No personal and/or family history of allergic diseases
  - Non-atopic
    - **Lack of positive skin tests to aeroallergens**
  - Normal serum levels of IgE
  - Examples: exercise induced asthma, aspirin intolerant asthma, obesity, and occupational exposure

Older categories: intrinsic asthma and extrinsic



Phenotypes of Idiosyncratic Asthma

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- **Exercise induced asthma**: caused by loss of heat, water, or both from lungs during exercise to the cooler or drier environmental air
  - Occurs within 10 minutes of starting, peaks after 20-30 minutes
  - May be a marker for poorly controlled asthma
  - In 11% of Olympic athletes
  - Diagnosis: exercise challenge test: ↑ HR to 80% of max for 4-6 mins
    - **Positive test**: ↓ in FEV1 (or PEF) after exercise
      - Measurements taken at 5 minute intervals for 20-30 minutes post-exercise
  - Treatment goal: participation in ANY physical activity without limitation
    - Longer warm-up periods are shown to be preventative
    - B2- agonists prevent >80% of exacerbations when used BEFORE activity



Phenotypes of Idiosyncratic Asthma

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- **Aspirin Intolerant Asthma (AIA) or Aspirin Exacerbated Respiratory Disease (AERD):** typically presents in adults within a three hour window of taking aspirin
  - Triad: asthma, nasal polyps and aspirin use
  - Symptoms 20 minutes to 3 hours after taking ASA
  - Estimated 5.5-12% of adults with asthma
- **Asthma with obesity**: elevated body mass index (BMI) and little eosinophilic airway inflammation
- **Occupational Exposure**: 5-10% of adult onset asthma
  - Variety of different exposures discovered; key to diagnosis is positive methacholine challenge test after exposure to specific agent



Etiology of Asthma

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A chronic heterogeneous disease of the airways. An individual with asthma will **always have** asthma but *will not always demonstrate* symptoms of an asthma.

Inflamma-tion of the lungs

Type 1 Hypersensitivity Reaction

Hyper-respon-siveness to stimuli

Airflow obstruction

reversible

Signs and symptoms of Asthma

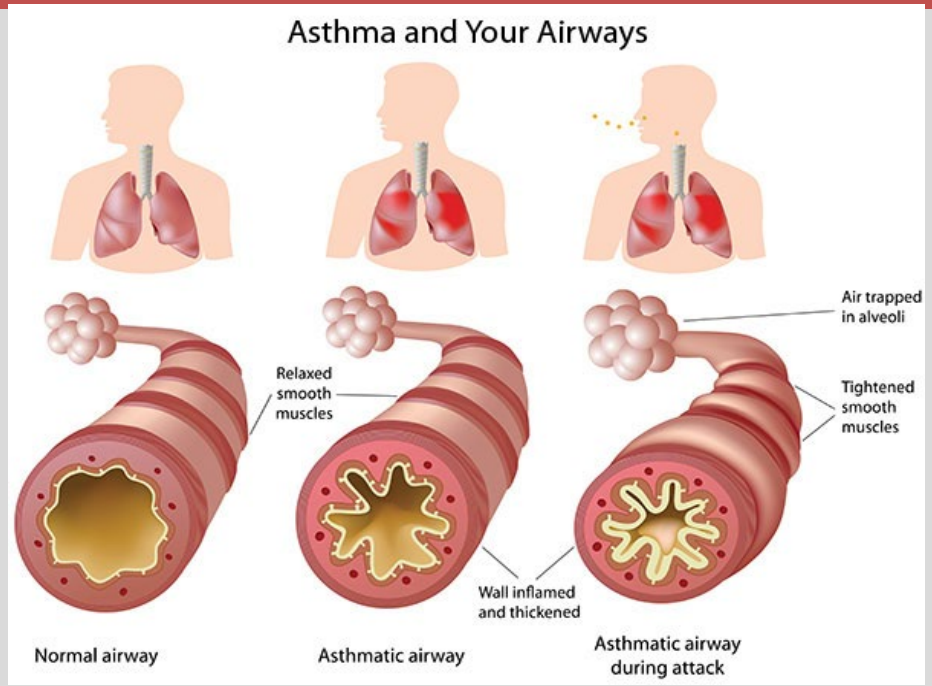
Asthma exacerbation

Wheezing
Breathlessness
Chest tightness
Nighttime or early morning cough

Etiology of Asthma

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An individual with asthma will **always have** asthma but *will not always demonstrate* symptoms of an asthma.





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Pathophysiology of Asthma: Airflow Obstruction

- The *smooth muscle dysfunction* with the increased **edema** and **mucous** lead to **obstruction** of the airway and symptomatic asthma:
  - Wheezing
  - Chest tightness
  - Cough
  - Shortness of breath
  - Trapping of CO2



Phases of Asthma

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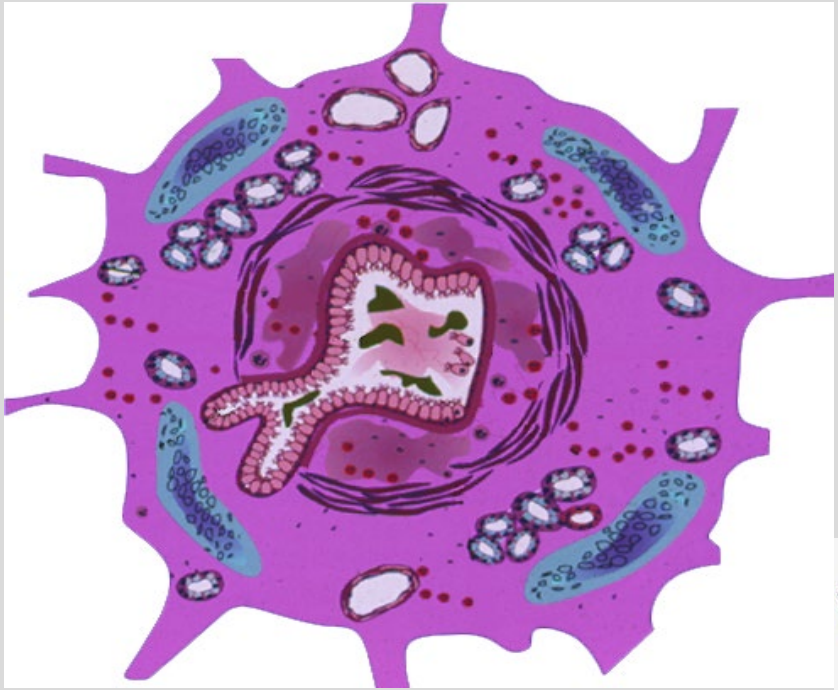
- Acute Response
  - Bronchial hyperactivity
  - Mucosal edema
  - Airway secretions
- Chronic Inflammatory Response
  - Increased inflammatory cell numbers
  - Epithelial damage
  - Airway remodeling



Chronic Changes of Asthma: Airway Remodeling

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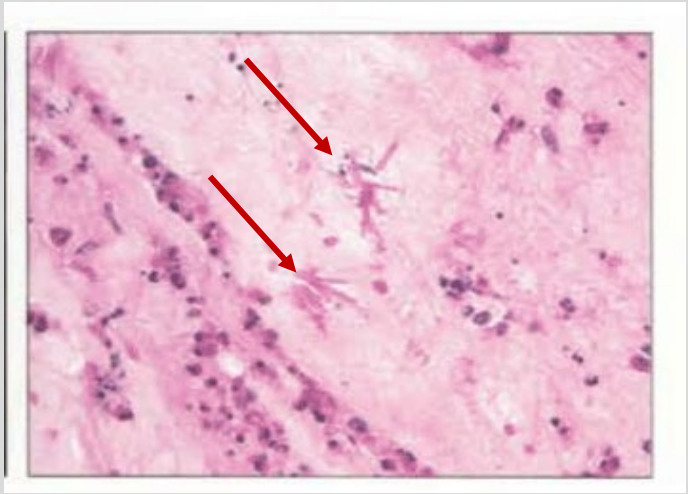
- Basement membrane thickening
- Vascular Dilation
- Edema (due to ↑ permeability)
- Inflammatory Cell Infiltration
  - Eosinophils, T-lymphocytes, neutrophils
- Epithelial Damage
- Smooth Muscle Hypertrophy
- Mucous gland hypertrophy and hyper secretion
- *Parenchyma is spared*



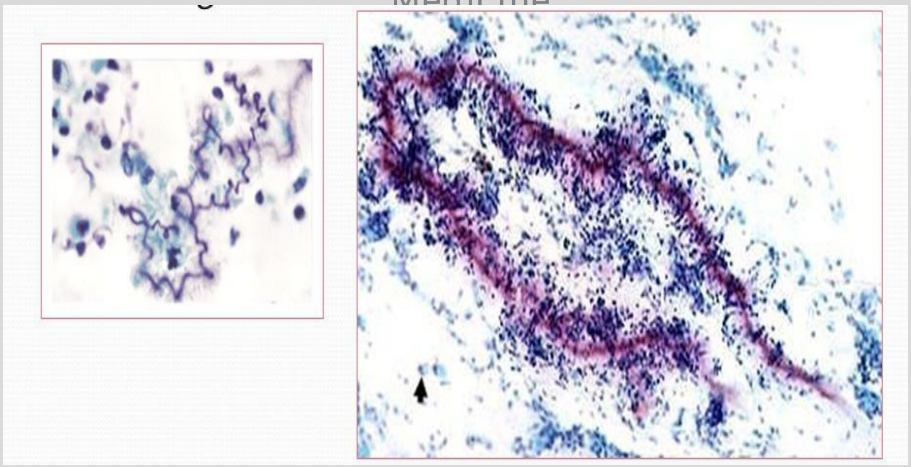


Sputum Pathology

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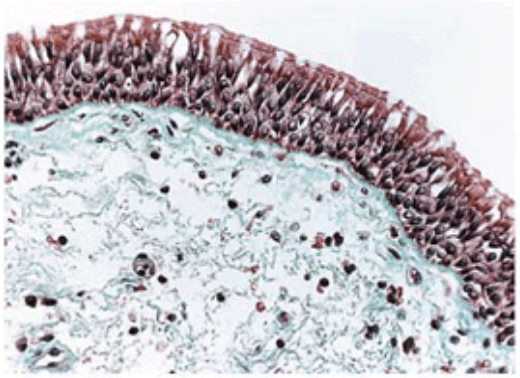


Charcot Leyden crystals are eosinophilic, double-pointed, needle-like crystals formed from breakdown of eosinophils in sputum.



Curschmann spirals are coiled, basophilic plugs of mucus formed in the lower airways and found in sputum and tracheal washings.





Subject without asthma

In the subject without asthma, the epithelium is intact; there is no thickening of the sub-basement membrane, and there is no cellular infiltrate.

Patient with mild asthma

In contrast, in the patient with mild asthma, there is evidence of goblet-cell hyperplasia in the epithelial-cell lining. The sub-basement membrane is thickened, with collagen deposition in the submucosal area, and there is a cellular infiltrate.

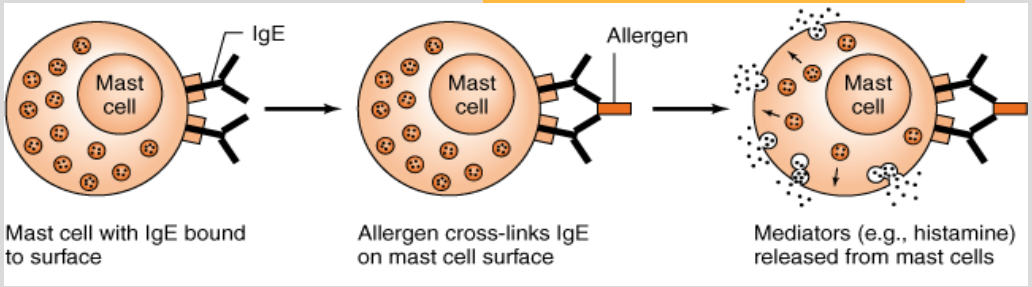


Risk Factors for Developing Asthma: Genetics

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- Strong familial component but incomplete understanding
- Genetic Links
  - Total serum IgE levels
  - Atopy
- Chromosomal abnormalities
  - 5q: cytokine clusters IL-4, IL-5, IL-9, IL-13 and β- adrenergic receptor, glucocorticoid receptor
  - 12q: codes for atopy and hyperresonspiveness

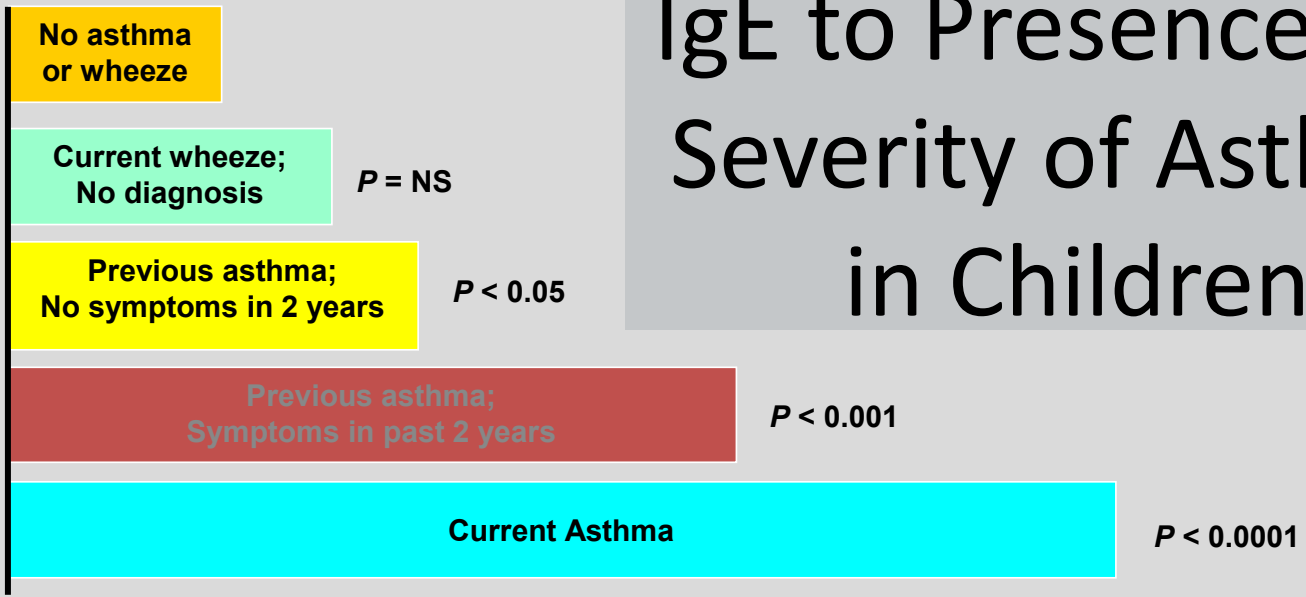
IgE and Asthma



- "Atopy, or the genetic predisposition to develop specific IgE antibodies directed against common environmental allergens, is the strongest identifiable risk factor for the development of asthma"
- IgE binds to high-affinity (FcεRI) receptors on mast cells and basophils and triggers the release of inflammatory mediators
- Increased IgE is associated with increased asthma prevalence and increased airway hyperresonspiveness
  - Low levels of IgE due not necessarily indicated lack of allergy
- Other causes for ↑ eosinophils NAACP (Neoplasia, Asthma, Allergy, Chronic adrenal insufficiency, Parasites [*invasive*])



Relation of Serum IgE to Presence and Severity of Asthma in Children

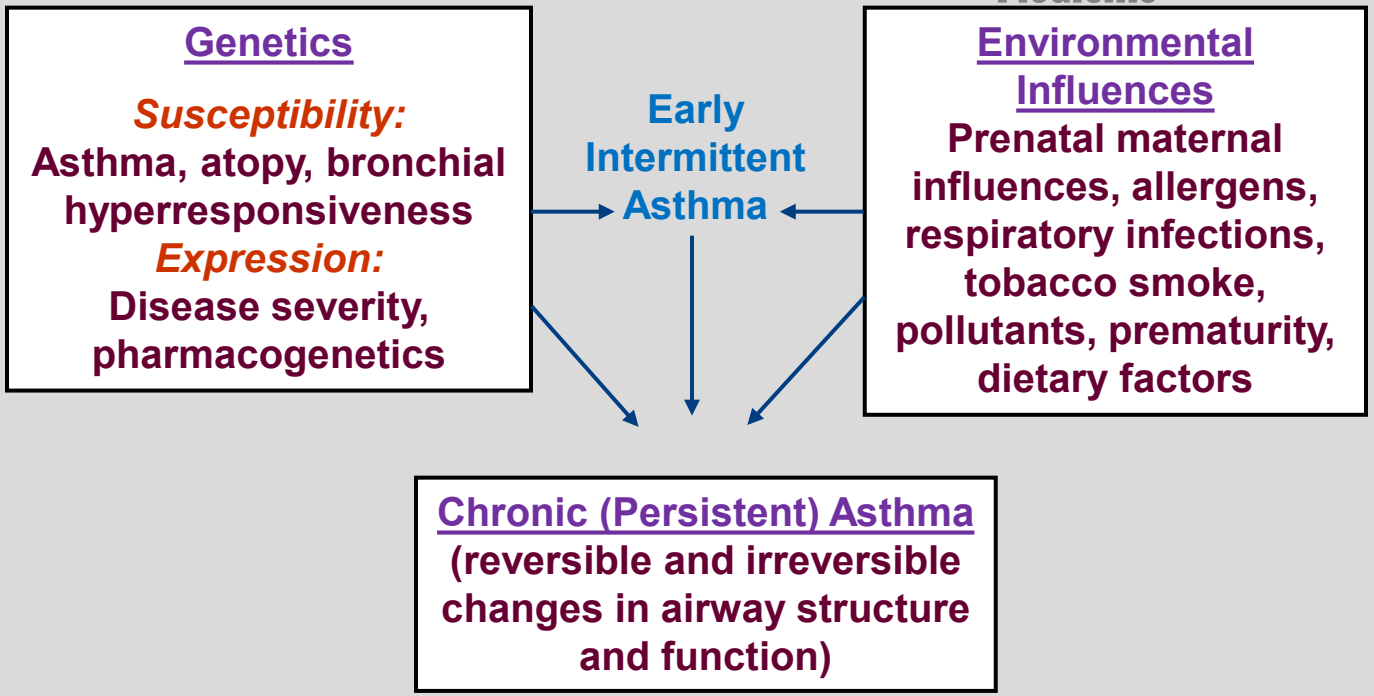


P values reflect comparison to group with no asthma or wheeze.

Sears MR et al. *N Engl J Med.* 1991;325:1067-71.

Gene-environment Interactions in Asthma Development

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Courtesy of E.R. Bleeker Jr., MD

Risk Factors for Asthma Development

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- **Biological Agents**
  - Sufficient evidence for causal relationship
    - **House dust mite**
  - Limited or suggestive evidence of association
    - Cockroaches (among pre-school aged children)
    - Respiratory syncytial virus (RSV)
- **Chemical Agents**
  - Sufficient evident of association
    - Environmental Tobacco smoke (among pre-school aged children)



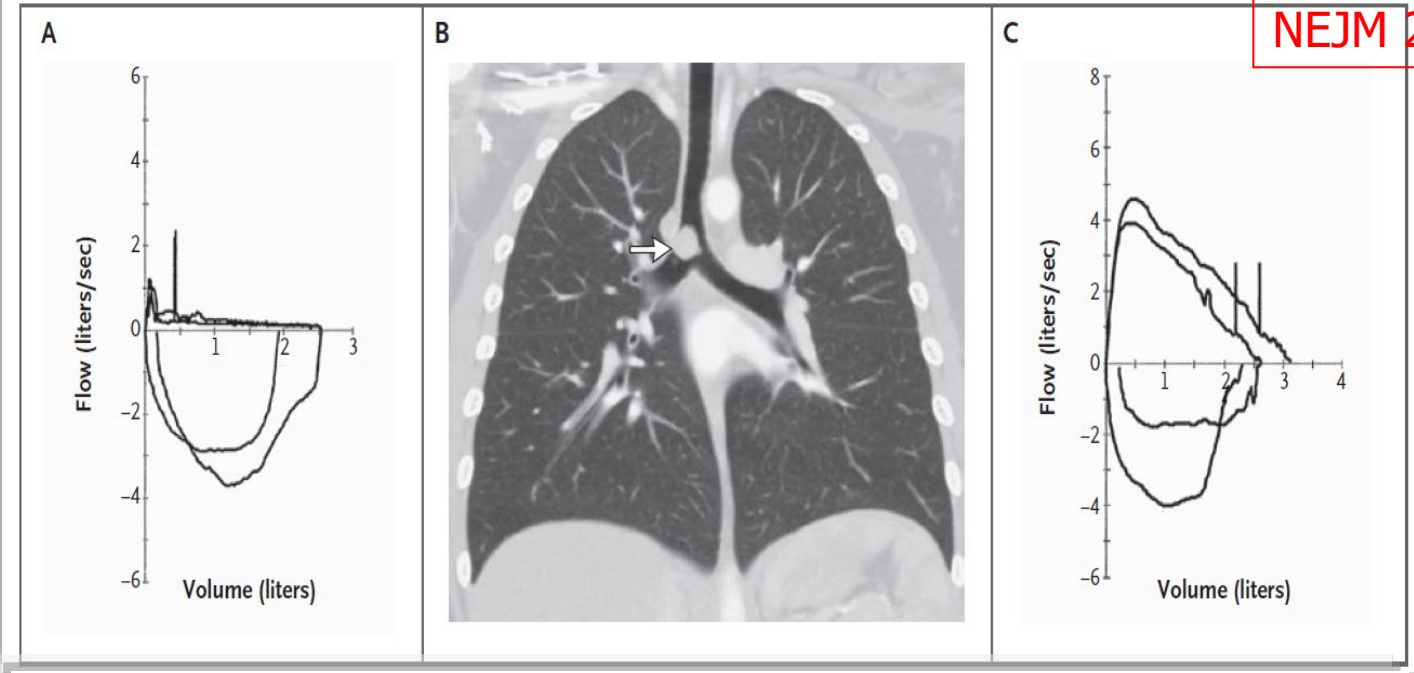
DIAGNOSIS

Diagnosing Asthma

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- No precise diagnostic test exists
  - "All that wheezes isn't asthma"
  - "Not all asthmatics wheeze"
- **To make the diagnosis you need**:
  1. Detailed medical history and physical that is suggestive of asthma
  2. Demonstration of variable expiratory airflow limitation
  3. Exclusion of alternative diagnoses

NEJM 2010



A 10-YEAR-OLD PREVIOUSLY HEALTHY GIRL PRESENTED WITH EPISODIC wheezing, dyspnea, and progressive exercise intolerance, which had developed during the preceding 9 months. On physical examination, the patient had moderate bilateral expiratory wheezing. She was treated for presumed asthma with multiple courses of inhaled bronchodilators, inhaled corticosteroids, montelukast, and oral corticosteroids. Though the patient reported that her condition had improved somewhat, there was no demonstrable objective improvement. Further evalu-

Diagnosing Asthma: History and Physical

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- **Symptoms of coughing, wheezing, SOB, chest tightness**
  - Coughing at night or after physical activity
  - Colds that last over 10 days
- **Pattern of symptoms**
  - Time of day? Time of year? With an exposure?
- **Severity of symptoms**
- Diagnosed with atopic dermatitis, eczema or other allergic skin conditions
- **Family history**
- **Relief when medication is used**

Diagnosing Asthma: History and Physical

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- **Physical Examination**
  - Wheezing
  - Tachycardia
  - Mild hypertension (systolic)
  - Nasal findings: polyps, edema, discharge,
  - Skin: eczema, atopic dermatitis
  - **Signs of SEVERE Airflow Obstruction**
    - Paradoxical pulse
      - ↓ in amplitude of systolic BP by >10 mmHg during inspiration; seen in cardiac tamponade, asthma, obstructive sleep apnea, pericarditis and croup
    - Hunched appearance
    - Use of accessory muscles
    - **Silent chest!**



Diagnosing Asthma: Testing Expiratory Airflow

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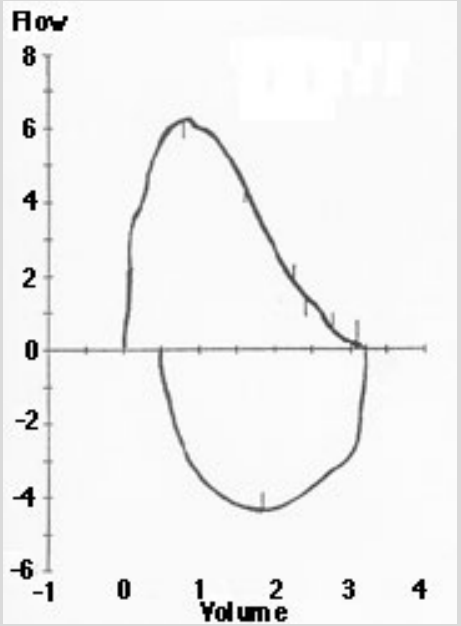
- **Spirometry**
  - A test used to measure the **VARIABILITY** lung function and to determine the expiratory airflow capacity
    - *Since asthma is a reversible airway obstruction it is key that this value changes during an acute asthmatic exacerbation and returns to baseline post-treatment*
  - FEV_1 (% of predicted normal)
    - At risk or mild ≥80%
    - Mildly reduced: 60-79%
    - Moderately reduced: 40-59%
    - Severely reduced: 30-49%
  - **This test by itself is NOT adequate for diagnosis**



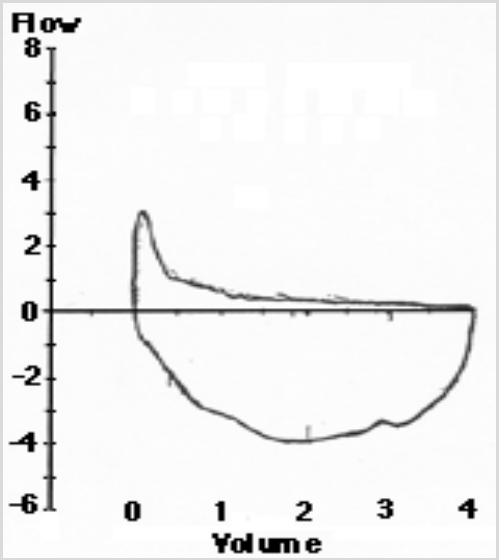
Diagnosing Asthma: Testing Expiratory Airflow

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- **Normal Flow Volume Loop**



- **Obstructive Disease Flow Volume Loop**
  - ***The SCOOP***



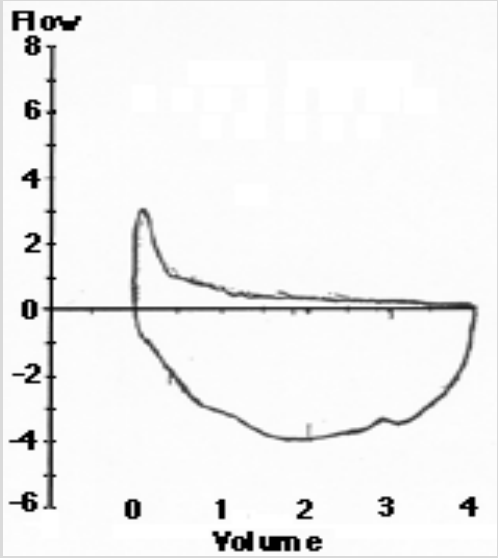


Diagnosing Asthma: Testing Expiratory Airflow

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- **Obstructive Disease Flow Volume Loop**
  - ***The SCOOP***



↓↓ FEV_1, ↓ FVC

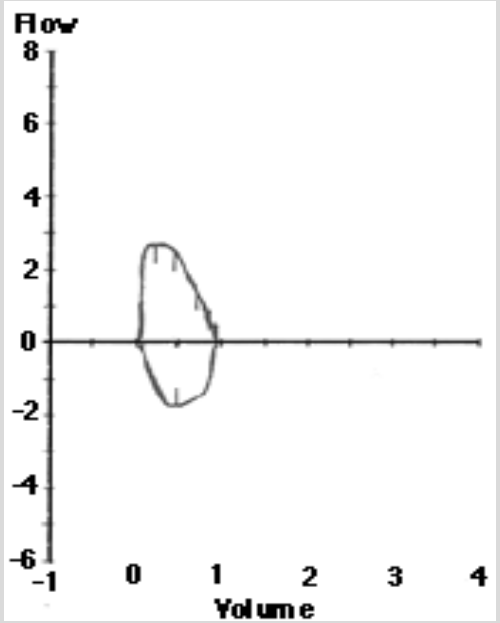
- This pattern is seen with:
  - Asthma
  - Chronic obstructive pulmonary disease (COPD)
    - Chronic bronchitis
    - Emphysema

The difference with asthma vs. COPD is that asthma is (typically) **REVERSIBLE**

Diagnosing Asthma: Testing Expiratory Airflow

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- **Restrictive Flow Volume Loop**



- **Pattern looks normal with a shift toward lower lung volumes and flow rates**
- *A reduced FVC with a normal FEV1/FVC is suggestive of a restrictive disorder*
  - Pulmonary fibrosis
  - *Pulmonary edema*
  - Interstitial pneumonia
  - *Diaphragm paralysis*



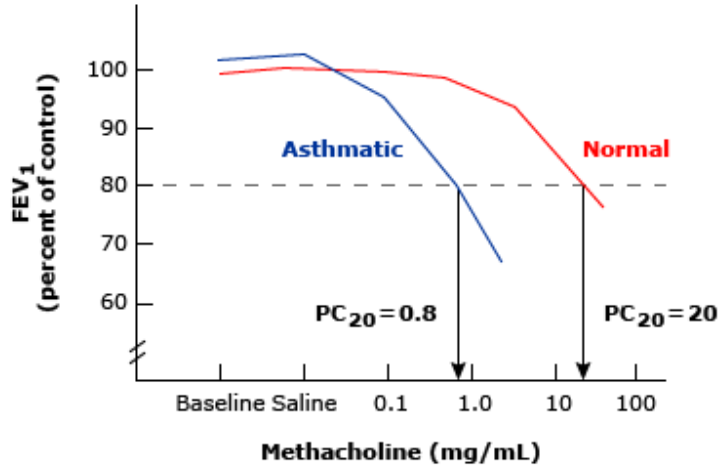
Diagnosing Asthma: Testing Expiratory Airflow

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- **Bronchoprovocation Challenge**
  - If a patient has the history suggestive of asthma and presents asymptomatic, this test can be implemented
    - Baseline FEV_1 is measured
    - Methacholine is inhaled to induce an acute exacerbation
      - Cholinomimetic agent; stimulates muscarinic receptors
    - FEV_1 is re-tested
      - **Positive test:** a decrease of 20% or more from baseline
    - The patient is treated with a bronchodilator
    - The patient is retested to show return of FEV1 to baseline



Diagnosing Asthma: Testing Expiratory Airflow



The effect of increasing the inhaled dose of methacholine in a healthy subject (red) and an asthmatic patient (blue). The provocative concentration is the amount of inhaled agonist required to drop the FEV_1 by 20 percent from the baseline (PC_{20} FEV_1) and is much less in the asthmatic than in the normal subject: 0.8 mg/mL versus 20 mg/mL. In general, a PC_{20} ≤8 mg/mL is consistent with asthma; and a PC_{20} >16 mg/mL is considered a negative test. Thus, an increase in airway responsiveness is characterized by a decrease in the PC_{20}.

UpToDate®

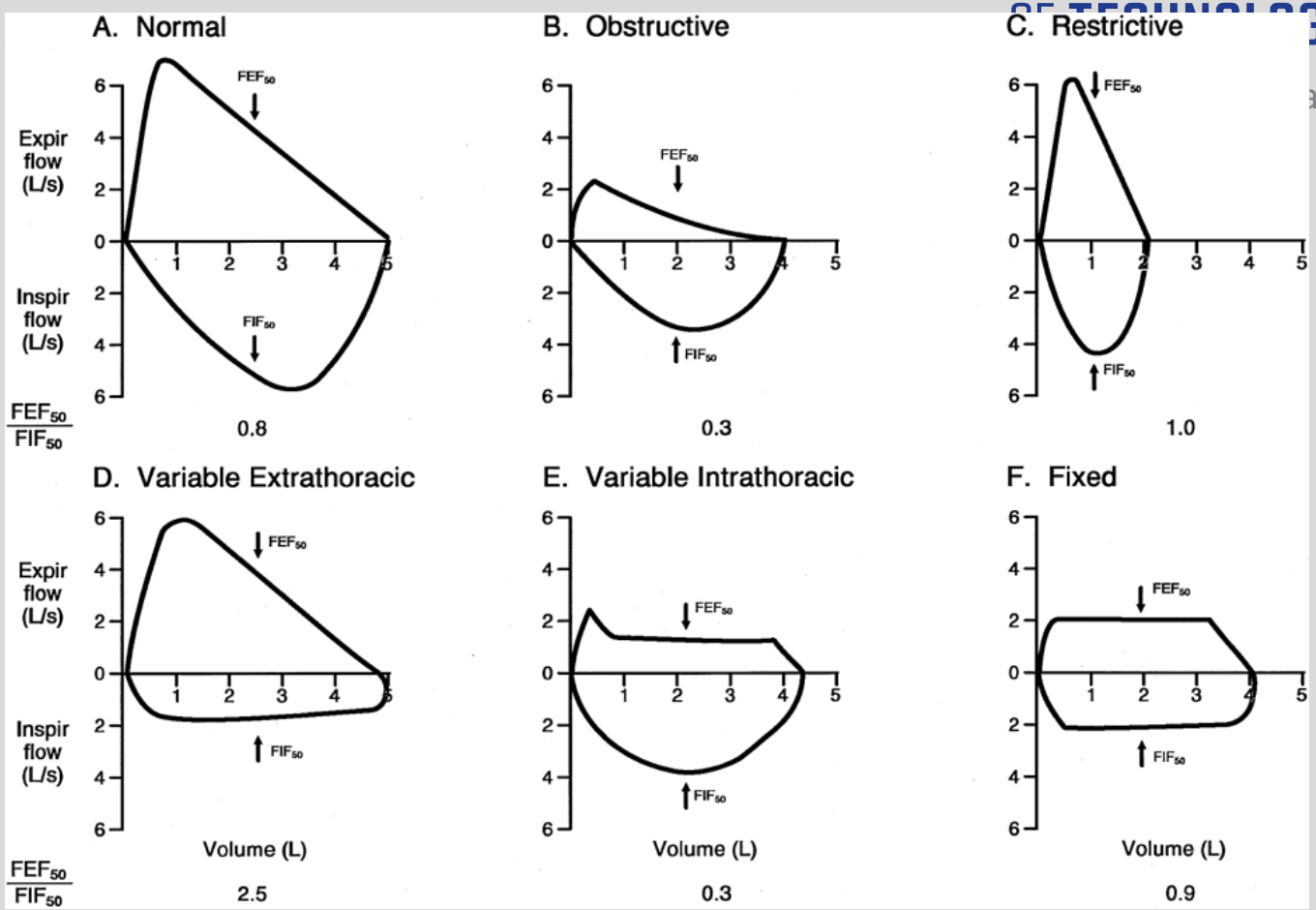


Diagnosing Asthma: Other

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- **Flow-Volume Loops**
  - Aids in detection of upper-airway obstruction
- **Lung Volume Measurements**
  - Usually normal
  - Air trapping seen during exacerbation
  - Vital capacity can be reduced
- **Airway Resistance**
  - Determined by plethysmography (body-box)
  - More sensitive measurement than FEV_1

The Major Patterns



Diagnosing Asthma: Other

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- **Diffusion Capacity**
  - Usually normal, may be elevated
    - Elevation indicated recruitment of capillaries from higher pulmonary arterial pressure
- **Hypoxemia** *RARE*
  - Usually indicated severe obstruction when present
  - Alveolar-arterial oxygen difference often elevated
- **Hypocapnia and respiratory alkalosis**
  - More common finding on arterial blood gas (ABG)
  - Normal (or increasing) pCO_2 during severe exacerbation indicated pending respiratory failure

Diagnosing Asthma: Excluding Differentials

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- **Infants and Children**
  - **Upper airway disease** (Allergic Rhinitis and Sinusitis)
  - **Obstruction involving large airways** (Foreign body, vocal cord dysfunction, vascular rings or laryngeal webs, Laryngotracheomalacia, tracheal stenosis, bronchostenosis, **Enlarged lymph nodes or tumor**
  - **Obstruction involving large airways** (Viral bronchiolitis or obliterative bronchiolitis, Cystic fibrosis, bronchopulmonary dysplasia, congenital heart disease)
  - **Other** (recurrent cough not due to asthma, aspiration from swallowing mechanism dysfunction or GERD)

Diagnosing Asthma: Excluding Differentials

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- **Adults**
  - **Chronic Obstructive Pulmonary Disease** (COPD)
  - **Congestive Heart Failure** (CHF)
  - **Pulmonary Embolism** (PE)
  - **Laryngeal dysfunction**
  - **Mechanical obstruction of the airways** (tumors)
  - **Pulmonary infiltrates with eosinophils**
  - **Cough secondary to drugs** (ACE inhibitors)
  - **Vocal cord dysfunction**

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Asthma Classification

Export to Powerpoint Print

Classifying asthma severity in youths greater than or equal to 12 years of age and adults

Components of severity		Classification of asthma severity (youths ≥12 years of age and adults)			
		Intermittent	Persistent		
			Mild	Moderate	Severe
Impairment Normal FEV_1/FVC: 8-19 yr 85 percent 20-39 yr 80 percent 40-59 yr 75 percent 60-80 yr 70 percent	Symptoms	≤2 days/week	>2 days/week but not daily	Daily	Throughout the day
	Nighttime awakenings	≤2x/month	3-4x/month	>1x/week but not nightly	Often 7x/week
	Short-acting $beta_2$-agonist use for symptom control (not prevention of EIB)	≤2 days/week	>2 days/week but not >1x/day	Daily	Several times per day
	Interference with normal activity	None	Minor limitation	Some limitation	Extremely limited
	Lung function	• Normal FEV_1 between exacerbations • FEV_1 >80 percent predicted • FEV_1/FVC normal	• FEV_1 ≥80 percent predicted • FEV_1/FVC normal	• FEV_1 >60 but <80 percent predicted • FEV_1/FVC reduced 5 percent	• FEV_1 <60 percent predicted • FEV_1/FVC reduced >5 percent
Risk	**Exacerbations requiring oral systemic corticosteroids**	**0-1/year (see footnote)**	**≥2/year (see footnote)**		
		Consider severity and interval since last exacerbation			
		Frequency and severity may fluctuate over time for patients in any severity category			
		Relative annual risk of exacerbations may be related to FEV_1			

Classifying severity for patients who are not currently taking long-term control medications. Level of severity is determined by assessment of both impairment and risk. Assess impairment domain by patient's/caregiver's recall of previous 2-4 weeks and spirometry. Assign severity to the most severe category in which any feature occurs. At present, there are inadequate data to correspond frequencies of exacerbations with different levels of asthma severity. In general, more frequent and intense exacerbations (eg, requiring urgent, unscheduled care, hospitalization, or ICU admission) indicate greater underlying disease severity. For treatment purposes, patients who had ≥2 exacerbations requiring oral systemic corticosteroids in the past year may be considered the same as patients who have persistent asthma, even in the absence of impairment levels consistent with persistent asthma.

PRECIPITATING FACTORS FOR ASTHMA EXACERBATIONS

Indoor Air Exposures & Asthma Exacerbations

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- **Biological Agents**
  - Sufficient evidence for causal relationship
    - Cat, Cockroach, and House Dust Mite
  - Sufficient evidence of association
    - Dog, Fungus/Molds, Rhinovirus
  - Limited or suggestive evidence of association
    - Domestic birds, Chlamydia, mycoplasma pneumonia, and RSV

- **Chemical Agents**
  - Sufficient evidence for causal relationship
    - Environmental Tobacco Smoke (among pre-school-aged children)
  - Sufficient evidence of association
    - NO_2, No_x (high levels)
  - Limited or suggestive evidence of association
    - Environmental Tobacco Smoke (among school-aged, older children, and adults), Formaldehyde, and Fragrances

Precipitating Factors for Asthma Exacerbations

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- Menstrual cycle and pregnancy
  - Premenstrual worsening
  - Effects of pregnancy unpredictable
    - **1/3 better**
    - **1/3 stay the same**
    - **1/3 worse**
      - Associated with carrying female fetus
- Medications
  - Aspirin and NSAID
  - β–adrenergic blockers
  - Sulfates, sulfites, metabisulfites

Precipitating Factors for Asthma Exacerbations

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- **Environmental Allergies**
  - Pollen, trees, gases, molds
- **Sleep**
  - Nocturnal symptoms common (3-4 AM)
    - Often overlooked because occurs so often
  - Also, early morning awakening
  - Endogenous circadian rhythms underlie nocturnal reduction in airway caliber
    - Decreased levels of catecholamines
    - Increased level of histamine

Precipitating Factors for Asthma Exacerbations

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- **Gastroesophageal reflux**
  - May exacerbate asthma symptoms
  - Vagally mediated reflexes involved
- **Sinus disease**
  - Persistent posterior drainage of nasal mucus
  - Sinusitis often found in patients with asthma

Global Initiative for Asthma 2024(GINA)

What's new in GINA 2024?

GINA 2024 update published 22 May 2024
Download from ginasthma.org

GINA Global Strategy for Asthma Management and Prevention

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The Global Initiative for Asthma (GINA)

- GINA was established by the WHO and NHLBI in 1993
  - To increase awareness about asthma
  - To improve asthma prevention and management through a coordinated worldwide effort
  - Independent since 2014, funded only by the sale and licensing of its reports and figures
- The GINA Strategy Report is a global evidence-based strategy that can be adapted for local health systems and medicine availability; it is widely used (downloaded from >200 countries)
- The GINA Strategy Report is updated every year
  - Twice-yearly cumulative review of new evidence (including GRADE reviews)
  - Evidence integrated across whole asthma strategy, not isolated PICOT questions
  - Careful attention to study design, populations, and clinical relevance
  - Extensive external review
  - Practical focus: not just 'what', but 'how'
- All members of GINA Science Committee are active in clinical asthma research
  - See www.ginasthma.com/about-us/methodology for handling of conflict of interest

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GINA guidance about COVID-19 and asthma

Updated 26 April 2021

GINA Global Strategy for Asthma Management and Prevention

www.ginasthma.org

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COVID-19 and asthma

- Are people with asthma at increased risk of COVID-19, or severe COVID-19?
  - People with asthma do not appear to be at increased risk of acquiring COVID-19, and systematic reviews have not shown an increased risk of severe COVID-19 in people with well-controlled, mild-to-moderate asthma
- Are people with asthma at increased risk of COVID-19-related death?
  - Overall, people with well-controlled asthma are not at increased risk of COVID-19-related death *(Williamson, Nature 2020; Liu et al JACI IP 2021)*
  - However, the risk of COVID-19 death was increased in people who had recently needed oral corticosteroids (OCS) for their asthma *(Williamson, Nature 2020)* and in hospitalized patients with severe asthma *(Bloom, Lancet Respir Med 2021)*.
- What are the implications for asthma management?
  - It is important to continue good asthma management (as described in the GINA report), with strategies to maintain good symptom control, reduce the risk of severe exacerbations and minimise the need for OCS
- Have there been more asthma exacerbations during the pandemic?
  - No. In 2020, many countries saw a *reduction* in asthma exacerbations and influenza-related illness. The reasons are not precisely known, but may be due to handwashing, masks and social/physical distancing that reduced the incidence of other respiratory infections, including influenza

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COVID-19 and asthma - medications

- Advise patients to continue taking their prescribed asthma medications, particularly inhaled corticosteroids (ICS)
  - For patients with severe asthma, continue biologic therapy or oral corticosteroids if prescribed
- Are ICS protective in COVID-19?
  - In one study of hospitalized patients aged ≥50 years with COVID-19, ICS use in those with asthma was associated with lower mortality than in patients without an underlying respiratory condition *(Bloom, Lancet RM 2021)*
- Make sure that all patients have a written asthma action plan, advising them to:
  - Increase controller and reliever medication when asthma worsens (see GINA report Box 4-2)
  - Take a short course of OCS when appropriate for severe asthma exacerbations
- Avoid nebulizers where possible, to reduce the risk of spreading virus
  - Pressurized metered dose inhaler via a spacer is preferred except for life-threatening exacerbations
  - Add a mouthpiece or mask to the spacer if required

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COVID-19 and asthma – infection control

- Avoid spirometry in patients with confirmed or suspected COVID-19, or if community transmission of COVID-19 is occurring in your region
  - Follow aerosol, droplet and contact precautions if spirometry is needed
  - Consider asking patients to monitor PEF at home, if information about lung function is needed
- Follow strict infection control procedures if aerosol-generating procedures are needed
  - Nebulization, oxygen therapy (including nasal prongs), sputum induction, manual ventilation, non-invasive ventilation and intubation
- Follow local health advice about hygiene strategies and use of personal protective equipment, as new information becomes available in your country or region

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TREATMENT

Treatment → PREVENTION

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- There is no cure for asthma
- The primary goal is **control of symptoms** through **reduction exacerbations** by limiting exposure of precipitating factors and medication
- When symptoms are not in control, a conversation must be had about exposures and use of medications.
  - Poor medication compliance is a major factor in uncontrolled asthmatics



Treatment → PREVENTION

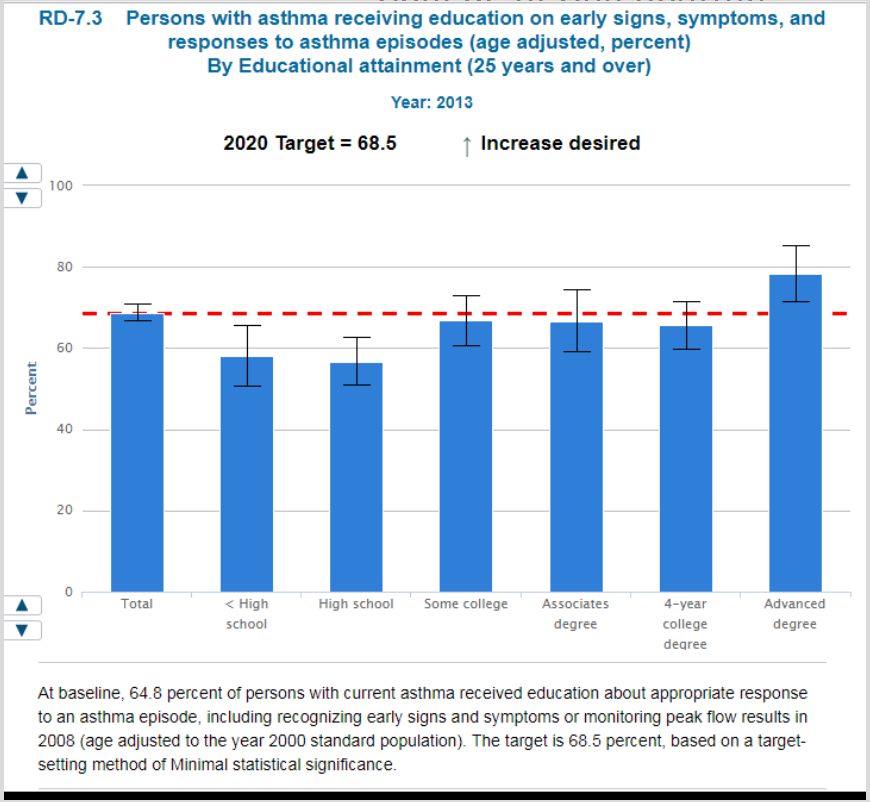
- Reduce house exposure to dust mites
  - Use bed encasements
  - Wash bed linens weekly
  - Limit stuffed animals to those that case be washed
  - Reduce relative humidity level (30-50% per EPR-3)
- Reduce exposure to environmental tobacco smoke
- Reduce exposure to cockroaches
  - Remove water and food sources
- Reduce exposure to molds
- Reduce exposure to pets
  - At a minimum do not allow pets in beds

Treatment → Education

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- The United State Task Force published a review article looking at a variety different levels of prevention measures (23 studies)
  - **Mild**: mattress and pillow covers
  - **Moderate**: teaching how to use bedding covers, household cleaning supplies, dehumidifiers, pest management and minor home structural repairs
  - **Major**: structural (remove carpet, replace ventilation, etc.)
- Economic Review: "The combination of minor to moderate environmental remediation with an educational component provides good value for the money invested based on improvement in symptom free days & savings from averted costs of asthma care and improvement in productivity."



Treatment → Education

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Checklist for uncontrolled asthma

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- A – allergy and adherence to therapy
- I – inflammation and infection
- R – rhinitis and rhinosinusitis
- E – exercise and error in diagnosis
- S – smoking
- M – medications
- O – occupational, hobby, and school
- G – gastroesophageal reflux

Courtesy of G. Volcheck, K. Lim, Mayo Clinic

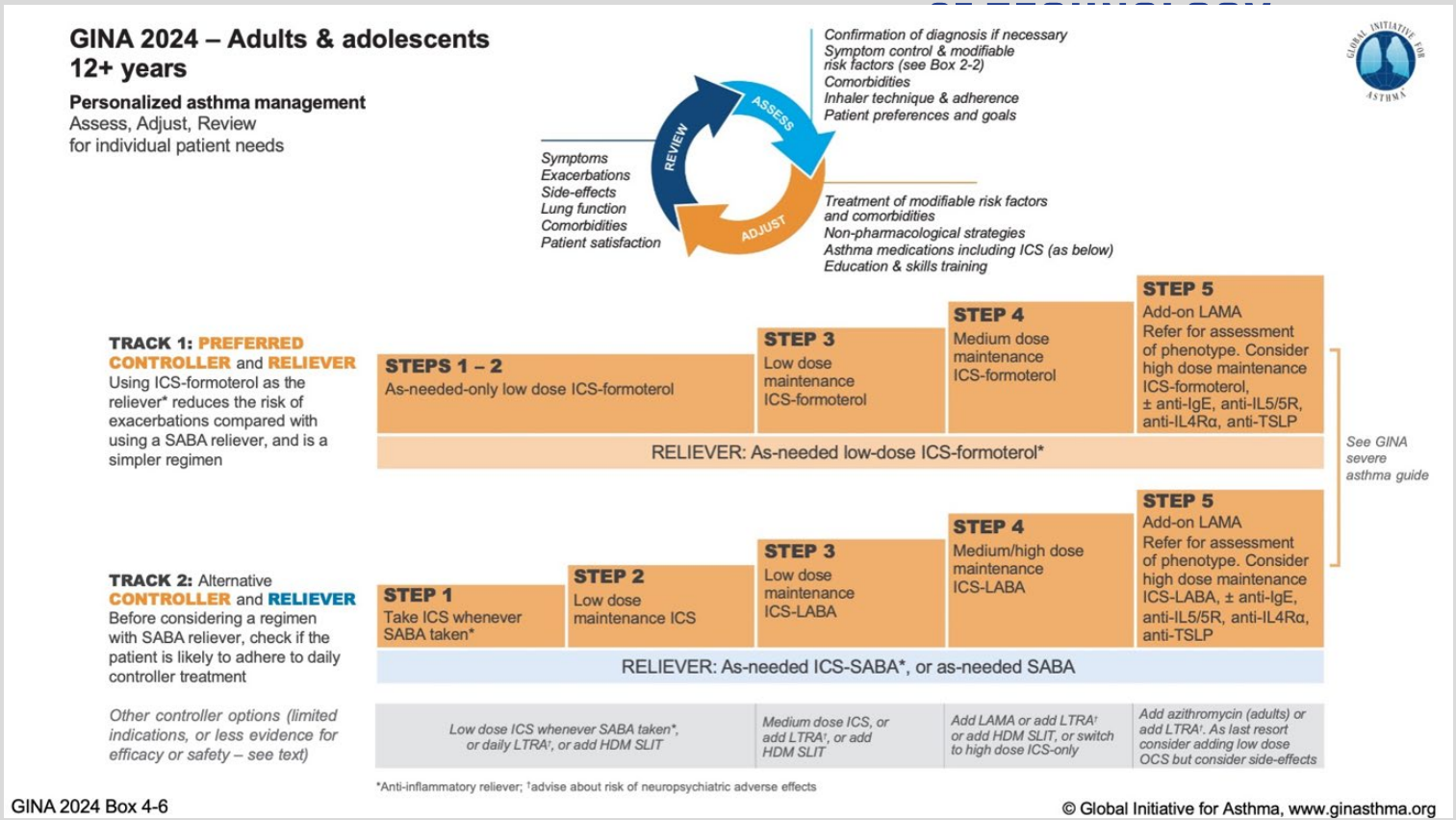
Medication

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- The step-wise approach is used for asthma control based on the severity of disease
- The specifics of the medications will be discussed in the pharmacology lecture

Medication: Step-Wise Approach





Medication Advancements

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- NEJM Article "Severe and Difficult-To-Treat Asthma in Adults"
  - A study to investigate how different phenotypes of asthma make symptom control more difficult
  - A broader understanding of the role of certain immune mediators are leading the way for new pharmacological discoveries
  - New pharmacological targets
    - Anti-IgE
    - Anti- IL-5 for circulating and for blocking interaction with IgE
    - Anti- IL-4 and anti-IL-13
    - Anti- CRTH2



Examples of Targeted Medications

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Drug Name	Biologic Mechanism of Action
Omalizumab	Anti-IgE; binds to Fc receptor of free IgE and reduces production of IgE
Mepolizumab	Anti-interleukin-5 (IL-5); binds circulation IL-5
Reslizumab	Anti-IL-5; binds circulating IL-5

Medications listed in the NEJM article "Severe and Difficult-To-Treat Asthma in Adults"

STATUS ASTHMATICUS

- An acute asthma exacerbation that does not respond to appropriate conventional treatment
- Implies a severe life-threating episode
  - Mortality is higher for minorities
- **Treatment**
  - IV corticosteroids
  - Inhaled β-2 agonists
  - ? IV $MgSO_4$
  - Heliox
  - Noninvasive ventilation (CPAP/BiPAP)
  - Mechanical ventilation



Summary Slide

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- Asthma is a chronic heterogeneous condition of the airway that presents with episodes of wheezing, breathlessness, chest tightness, nighttime or early morning cough
- Asthma has a significant impact on activities of daily living
  - Effects more African American people than other races
  - Disproportionally effects people below the poverty line and people living in inner cities
  - Boys more than girl; women more than men
- Asthma is classified as allergic or idiosyncratic (non-allergic)
  - Allergic is associated with higher levels of IgE and atopy
  - IgE stimulates mast cells to release histamine, cytokines and leukotrienes leading to the symptoms of asthma
- The environment and genetics play a key role in the development of asthma
- Inflammation + hyperresonspiveness → airway obstruction → asthma symptoms

Summary Slide

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- **Diagnosis**:
  1. Detailed medical history and physical that is suggestive of asthma
  2. Demonstration of variable expiratory airflow limitation
     - "the scoop", ≥ 20% decrease with methacholine challenge, reversible!
  3. Exclusion of alternative diagnoses
- Status asthmatics is defined by an acute asthma exacerbation that FAILS to respond to appropriate treatment
  - Red flags: "silent chest", rise in pCO_2, drowsiness, cyanosis, and failure to improve with treatment
- The goal of asthma treatment is CONTROL by prevention of acute exacerbations
  - Avoiding irritants, medication compliance
  - Education is key to preventing acute asthma attacks

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R